Can a Fisher/Cramér-Rao Bound Predict θ's Empirical Recovery Scatter?

Full-covariance CRB vs. real pixel-likelihood fits, aberrated wavefront

Contents

1 Motivation

Looking at 2D residual plots of $\hat{\theta}_{\text{best}} - \theta_{\text{true}}$ for the `confocal_random_zernike` (aberrated-wavefront) dataset, the scatter looks Gaussian. That raises the obvious question: can a Fisher-information / Cramér-Rao bound predict that spread analytically, rather than only measuring it empirically?

This was tried once before and got stuck. From `notes/joint_profile_inference.tex` (a sibling investigation, §2026-07-20): a Laplace approximation using the *diagonal* of a local finite-difference Hessian at θ's MAP was found **66–220$\times$ tighter** than the true empirical scatter across 50 shots. That work diagnosed why, and flagged two things that were never actually done:

1. θ's failure mode is a *ridge* – a continuous, weakly-identified direction (near-zero Fisher eigenvalue), not discrete multimodality like ϕ's. A ridge is compatible with local Fisher/CRB validity *if* it is close to linear/quadratic near the truth – but whether it actually is was flagged as unchecked (Future Work item 0a in that file) and never verified.
2. The failed attempt used a **diagonal-only** Hessian, discarding exactly the off-diagonal correlations a ridge produces. `crb_signal.py` already has `build_joint_fisher()` (full 17×17 joint Fisher over $(\phi, \theta_{z0}, \theta_{z100})$, pixel-likelihood mode, all cross terms) and `full_per_shot_covariance()` (inverts the *full* matrix) – but the latter was dead code, never called from that module's own driver. Nobody had tried the full-covariance version.

This note finishes that specific, previously-flagged, previously-skipped check, reusing the existing (but previously unwired) machinery rather than re-deriving anything. Scope, by explicit direction: **one dataset** (the aberrated `confocal_random_zernike` case), matching CRB to empirical data. A Gaussian-beam-vs-aberrated comparison, and the downstream connection to the β (signal) CRB, are deliberately deferred.

2 Setup

Dataset: `data/R20_N200_A1000000_..._psmapconfocal_random_zernike` (20 runs $\times$ 200 shots, 10^6 atoms/shot, aberrated wavefront via `phase_space_grids.py`'s interpolated-PSMAP path). Cloud prior: independent Gaussian on each of the 8 kinematic components (position, velocity, position spread, velocity spread, in x and y), $\sigma = 10\,\mu$m (or the matching velocity unit) for all 8. Evaluator geometry: **exactly** what `non_phase_shear/analysis/generate_kinematic_estimates.py`'s `best` (pixel-likelihood) fit uses – `bins=32`, tight half-range 1.54×10^{-3} m, `pixel_n_gh=8` – so the CRB is evaluated on the identical pixel/bin configuration that produced the empirical numbers it is compared against. Using a different geometry would make the comparison apples-to-oranges before it starts.

All CRB quantities below are evaluated at the *true* $(\phi, \theta_{z0}, \theta_{z100})$ for each shot – the standard Cramér-Rao convention, and also convenient here since it needs no extra optimizer run (the empirical $\hat{\theta}_{\text{best}}$ values already come from a separate, already-run fit).

Implementation: `python-scripts/theta_crb_vs_empirical.py` (new), importing `crb_signal.py`'s Fisher-matrix code rather than duplicating it.

3 Step 1: Is the ridge direction actually linear/quadratic?

For 10 real shots (`run_000`), built the full 17×17 joint Fisher matrix H at the true $(\phi, \theta_{z0}, \theta_{z100})$, found its smallest eigenvalue $\lambda_{\min}$ and eigenvector v (the ridge direction; $\sigma_{\text{ridge}} = 1/\sqrt{\lambda_{\min}}$), and profiled the *real* Poisson NLL along v at $k\sigma_{\text{ridge}}$ for $k \in \{0.5, 1, 2, 4, 8\}$, comparing to the Fisher-implied quadratic $\frac{1}{2}\lambda_{\min}(k\sigma_{\text{ridge}})^2$.

Result: not cleanly quadratic. The ratio (actual ΔNLL / quadratic ΔNLL) at $k = 0.5$–1 (the scale that actually matters – real fluctuations live near 1σ, not 8σ) ranged from -2.9 **to** $+4.3$ across the 10 shots – including several shots where the sign itself flips (the true NLL is briefly *lower*, not higher, than at the anchor point, in the direction the Fisher matrix calls "increasing"). The ratio does converge toward 1 at larger k (e.g. 0.80–1.23 at $k = 8$), consistent with the quadratic approximation eventually dominating far from the anchor, but that is not the regime real shot noise samples.

This is a genuine, unresolved finding, not swept under the rug: the single worst-conditioned direction is *not* well-approximated by the local quadratic at the scale that matters. And yet – see §4 – the *aggregate*, full 8-dimensional marginal covariance built from the same local Fisher matrix matches the empirical scatter well. The most likely reconciliation: v (found by minimizing over the full 17-dimensional $(\phi, \theta_{z0}, \theta_{z100})$ space) is a mixed direction, not aligned with any single physical θ component's own marginal-variance direction, so a 2–4× local mis-estimate of curvature along v does not necessarily translate into a comparable error in any individual component's marginal variance after the Schur-complement/inversion mixes all 17 directions together. This is a hypothesis, not something checked here – flagged in §8 as follow-up.

4 Step 2+3: Full-covariance CRB vs. empirical RMSE

For the same 10 shots, computed `full_per_shot_covariance(H)` and extracted $\sigma_{\theta_{z0}}$, $\sigma_{\theta_{z100}}$ (marginal std per component), aggregated as root-mean-square across shots. Compared against the empirical RMSE of $\hat{\theta}_{\text{best}} - \theta_{\text{true}}$, pooled over z0 and z100, from the 13 completed runs of the real `confocal_random_zernike` job available at the time of this check (5200 shot/z-slice points).

One data-quality issue found and handled: the raw empirical RMSE for σ_{vy} was 2.33×10^{-6}, a 3.46× mismatch against the CRB's 6.75×10^{-7} – while every other component matched to a few percent. Investigated directly rather than accepted: of 5200 points, a single shot has a σ_{vy} residual of 1.6×10^{-4} (161 μm) – roughly 70× larger than the next-largest residual in that component and wildly inconsistent with the rest of the distribution (median |residual| = 4.4×10^{-7}, matching σ_{vx}'s median almost exactly). RMSE is quadratic in residuals, so this one likely optimizer failure (an L-BFGS-B run landing in a bad basin for that one shot – not a CRB or model problem) was enough to dominate the whole-sample RMSE by itself. Excluding it (a MAD-based robust-outlier filter, $> 20\sigma_{\text{robust}}$ in any component; 1/5200 points excluded) brings σ_{vy}'s empirical RMSE to 6.77×10^{-7}.

Component	CRB (avg $z0$,$z100$)	Empirical RMSE (outlier excl.)	Ratio
μ_{x0}	9.655×10^{-6}	9.816×10^{-6}	1.017
μ_{y0}	9.655×10^{-6}	9.531×10^{-6}	0.987
μ_{vx0}	2.471×10^{-6}	2.511×10^{-6}	1.016
μ_{vy0}	2.473×10^{-6}	2.437×10^{-6}	0.985
σ_x	9.500×10^{-6}	9.912×10^{-6}	1.043
σ_y	9.461×10^{-6}	9.892×10^{-6}	1.046
σ_{vx}	6.767×10^{-7}	6.861×10^{-7}	1.014
σ_{vy}	6.745×10^{-7}	6.771×10^{-7}	1.004

Table 1: Full-covariance theta CRB vs. empirical RMSE (outlier-excluded), all 8 kinematic components, confocal_random_zernike dataset. All values in metres (or the matching velocity unit). Every component agrees to within ±5%.

Result: the full-covariance CRB matches the empirical scatter to within a few percent, across all 8 components, once one known bad fit is excluded. This directly refutes the presumption (reasonable, given the prior 66–220× failure) that Fisher/CRB simply cannot predict θ's recoverability – it was never actually tried with the full covariance before now.

5 Extension: the unaberrated (Gaussian-beam) baselines

The above was scoped to the aberrated dataset only; this section repeats §4's CRB-vs-empirical comparison (Step 2+3 only, not the ridge check) for the two already-fit, unaberrated (`CONFOCAL_FINE`) baseline datasets already saved in `non_phase_shear/results/` (`kinematic_estimates_{1e6,1e8}_N10_shots50`
– no new data generation, reusing `python-scripts/theta_crb_gaussian_baseline.py`.

Data-availability caveat for `1e6`. Its source dataset is no longer on disk (the only remaining 1e6-atom, 50-shots/run directory has a different cloud-width prior than what the saved `true_theta` values actually show, so it isn't the same source). The saved JSON has `true_theta` and photon-count totals (n_{tot}) – both of which `build_joint_fisher` needs – but not the true ϕ. Since ϕ is drawn independently of θ (`phi0_mode=random`), this uses a random $\phi \sim U(0, 2\pi)$ substitute per shot instead. `1e8` has its exact source data (`data/R40_N50_A100000000_..._f0.3000`, confirmed by matching `true_theta`/photon-count against its own file attrs), so its numbers use the real per-shot ϕ, no substitution.

Component	CRB (1e6)	Empirical (1e6)	Ratio
μ_{x0}	9.644×10^{-6}	1.017×10^{-5}	1.054
μ_{y0}	9.644×10^{-6}	9.676×10^{-6}	1.003
μ_{vx0}	2.500×10^{-6}	2.638×10^{-6}	1.055
μ_{vy0}	2.501×10^{-6}	2.499×10^{-6}	0.999
σ_x	9.389×10^{-6}	1.019×10^{-5}	1.086
σ_y	9.367×10^{-6}	9.990×10^{-6}	1.067
σ_{vx}	6.676×10^{-7}	7.053×10^{-7}	1.056
σ_{vy}	6.699×10^{-7}	6.909×10^{-7}	1.031

Table 2: `1e6` baseline (unaberrated, ϕ substituted – see caveat above): matches to within ±9%, similar quality to the aberrated result.

Component	CRB (1e8)	Empirical (1e8)	Ratio
μ_{x0}	7.616×10^{-6}	9.279×10^{-6}	1.218
μ_{y0}	7.656×10^{-6}	9.295×10^{-6}	1.214
μ_{vx0}	1.972×10^{-6}	2.402×10^{-6}	1.218
μ_{vy0}	1.982×10^{-6}	2.408×10^{-6}	1.215
σ_x	6.262×10^{-6}	9.877×10^{-6}	**1.577**
σ_y	6.304×10^{-6}	9.822×10^{-6}	**1.558**
σ_{vx}	4.297×10^{-7}	6.703×10^{-7}	**1.560**
σ_{vy}	4.205×10^{-7}	6.731×10^{-7}	**1.601**

Table 3: `1e8` baseline (unaberrated, real ϕ, no substitution): a real, non-trivial gap – means off by ∼21%, widths off by ∼56–60%. Confirmed this is not a numerical artifact: the CRB is stable under h_θ from 10^{-6} to 10^{-8} (only degrading at 10^{-9}, floating-point roundoff), and the empirical residual distribution is broadly heavy-tailed (max ≈ 4× median, consistent across all 8 components) rather than dominated by one outlier shot as in §4.

Interpretation. `1e6` (both aberrated and unaberrated) and the aberrated case both match the CRB to a few percent. `1e8` does not – empirical scatter is systematically 1.2–1.6× the CRB, worst for the width parameters. This is not swept under the rug, and a plausible explanation already exists in prior work rather than needing to be invented here: `notes/joint_profile_inference.tex` (§2026-07-20, a different optimizer but the same underlying pixel likelihood) documented that θ's optimisation *"genuinely struggles at* 10^8*"* – even warm-started at the exact true θ, a conditional solve "wandered/ plateaued... never approaching tolerance" – attributed to the likelihood surface becoming legitimately much sharper at high photon count ($|\nabla_\theta \mathrm{NLL}|$ exceeding the NLL itself). If `fit_theta_best`'s L-BFGS-B is similarly not fully converging at 10^8 (not checked here – see §8), that would inflate the *empirical* RMSE above the true CRB floor without the CRB itself being wrong – consistent with the direction of the observed gap (empirical > CRB, not the reverse) and with 1e6 (where the same prior work reported convergence was fine) matching cleanly.

6 Framing: this is a single-shot comparison, and that's intentional

θ is a fresh, independent nuisance parameter drawn from the prior *every shot* – it is not a shared, estimated-once parameter like $\beta = (A_s, A_c)$. Consequently, pooling more shots does *not* reduce the true per-shot θ RMSE (there is nothing to pool: shot i's θ_i tells you nothing about shot j's θ_j). What more shots *do* improve is the *measurement precision of the empirical RMSE estimate itself* – with 5200 points the empirical RMSE's own relative uncertainty is $\sim 1/\sqrt{2 \times 5200} \approx 1\%$, tight enough to distinguish the observed few-percent CRB agreement from noise. This is why this comparison uses many shots (for a trustworthy *measurement*) while still being a fundamentally single-shot statement (the CRB itself is a per-shot quantity, not one that shrinks with N_{shots}) – unlike β's CRB, which genuinely does improve with shot count (`crb_signal.py`'s `crb_from_j`/`simple_scaling_crb`).

7 From θ-CRB to β-CRB

7.1 Method: propagating theta uncertainty into a phase-information number

`crb_signal.py`'s `per_shot_information` is exactly the theta-uncertainty-propagation step: it takes the same joint 17×17 Fisher matrix H used in §4 (over $\phi, \theta_{z0}, \theta_{z100}$) and Schur-complements *only* the 16 theta directions out (not ϕ), leaving a scalar effective per-shot phase information j_{eff} – literally "how much information about the laser phase ϕ_i survives, after accounting for the fact that θ is unknown and must be estimated from the same image." Since $\delta\phi_i(\beta) = A_s \sin(2\pi f i) + A_c \cos(2\pi f i)$ enters only through ϕ_i's argument at $z100$, the N-shot Fisher matrix for $\beta = (A_s, A_c)$ is exactly $F_\beta = \sum_i j_i \, x_i x_i^\top$, $x_i = [\sin(2\pi f i), \cos(2\pi f i)]$ (`crb_from_j`), and $\mathrm{CRB}(\beta) = F_\beta^{-1}$.

7.2 Converting to rad/$\sqrt{\text{shot}}$ and rad/$\sqrt{\text{Hz}}$

Two useful normalisations, both already implemented:

- **rad/$\sqrt{\text{shot}}$** (a shot-count-independent number). When shots densely sample many signal periods ($\langle x_i x_i^\top \rangle \to \frac{1}{2} I_2$, true here: $f = 0.3$ cyc/shot over 50-200 shots is 15-60 periods), $F_\beta \approx (N j_{\mathrm{bar}}/2) I_2$, so $\sigma_{A_s} = \sigma_{A_c} \approx \sqrt{2/(N j_{\mathrm{bar}})} = \sigma_{1\mathrm{shot}}/\sqrt{N}$ (`simple_scaling_crb`). $\sigma_{1\mathrm{shot}} = \sqrt{2/j_{\mathrm{bar}}}$ is what "rad/$\sqrt{\text{shot}}$" means: it is the number such that the CRB at any shot count N is $\sigma_{1\mathrm{shot}}/\sqrt{N}$ – report this number once, and it answers "what precision at N shots" for any N without recomputing anything.

- **rad/$\sqrt{\text{Hz}}$** (an amplitude spectral density, the standard sensitivity unit in precision metrology). This requires one more piece of information this codebase does not define anywhere: the real shot *repetition rate* f_{rep} (shots/second) – every frequency in this codebase (`--signal_freq`) is in cycles/*shot*, not Hz, so there is no absolute time axis without it. Given f_{rep}, standard white-noise-ASD conversion applies:

$$\text{ASD [rad}/\sqrt{\text{Hz}}] \;=\; \sigma_{1\text{shot}}\ [\text{rad}] \times \sqrt{T_{\text{rep}}} \;=\; \sigma_{1\text{shot}}/\sqrt{f_{\text{rep}}}, \qquad T_{\text{rep}} = 1/f_{\text{rep}}.$$

(Same logic as converting a per-sample noise std to a PSD/ASD for any white, sample-limited process: divide by $\sqrt{\text{sample rate}}$. Valid here because each shot's θ_i, ϕ_i are drawn independently – no shot-to-shot correlation to worry about.) $\sigma_{1\text{shot}}$ is reported below; multiply by your own f_{rep} whenever you have one.

Implementation: `python-scripts/beta_crb_vs_empirical.py`. Uses a representative sample of shots (not all N) to estimate j_{bar} (checked: relative shot-to-shot spread of j is ∼9% for `1e6`, tight enough that treating $j_i \approx j_{\text{bar}}$ for all N shots when building F_β via the real x_i pattern is a good approximation, while avoiding an $O(N)$-shot Fisher-matrix cost).

7.3 Result: `1e6` (Gaussian beam)

	$\sigma_{1\text{shot}}$ [rad]	CRB(N=50)	Empirical (N=50, `best`)
A_s	1.14×10^{-3}	1.62×10^{-4}	4.95×10^{-4}
A_c		1.62×10^{-4}	3.21×10^{-4}

Table 4: $j_{\text{bar}} = 1.53\times10^6$ (rel. shot-to-shot spread 9%), 30 sample shots. Ratio (empirical/CRB): 3.05× (A_s), 1.99× (A_c).

Unlike θ's own CRB (which matched to ±5%), β's CRB under-predicts the empirical scatter by a real, substantial factor. Checked and ruled out one concrete hypothesis before accepting this: rebuilt the CRB using *exactly* `beta_fits_from_kinematics.py`'s own evaluator geometry (`bins=16`, native half-range, vs. the tight/32-bin geometry used elsewhere in this note) – j_{bar} changed by $< 0.3\%$ (1.523×10^6 vs. 1.526×10^6), so evaluator geometry is *not* the explanation.

The actual explanation is structural. `beta_fits_from_kinematics.py`'s `best` method is a *two-stage plug-in* estimator, not the idealised joint estimator this CRB models:

1. Fit $\hat{\theta}_{\text{best}}$ first (a point estimate – whose own scatter matches its CRB, §4).
2. Convert to `crb_signal.py`'s η representation via `theta_to_eta`, which **hardcodes two cross-terms to zero**: `[mu_x0, mu_vx0, mu_y0, mu_vy0, sx**2, 0.0, svx**2, sy**2, 0.0, svy**` – exactly the position-velocity covariance terms.
3. Fit β from that fixed, cross-term-stripped η – θ's uncertainty has already been collapsed to a point estimate and its correlation structure discarded, not properly marginalised.

This CRB assumes an efficient estimator that uses the full joint (ϕ, θ, β) information optimally (Fisher-matrix elimination is the textbook-correct way to get the marginal CRB for *an efficient* estimator). The real pipeline throws away exactly the correlation structure θ was shown to actually have (§3's ridge) partway through. A two-stage plug-in estimator falling short of the joint bound is a standard, expected effect, not a bug in either number – the CRB is correctly a lower bound; this particular real pipeline doesn't saturate it.

`1e8` and the aberrated dataset are pending (see §8) – this section will be extended once the phase-shear-readout comparison (also requested, separate note or section) is done.

8 Open questions / future work

- **Reconcile §3 and §4.** Check what the worst-eigenvalue direction v actually projects onto in θ-component space, for a few shots – confirm (or refute) the "mixed direction, doesn't dominate any single marginal" hypothesis directly rather than leaving it as speculation.
- **Diagnose the 10^8 gap** (§5): check whether `fit_theta_best`'s L-BFGS-B is actually converging (`res.nit`/`res.success`, and/or a tighter `maxiter` / warm-start-from-truth spot check) at 10^8 atoms specifically. If convergence is the cause, this is an estimator-quality issue, not a CRB-validity issue – worth confirming directly rather than resting on the plausible-but-unverified analogy to the joint-solve's documented 10^8 struggles.
- **Exact (not ϕ-substituted) `1e6` baseline** and a dedicated apples-to-apples aberrated-vs-Gaussian comparison at matched N: the `1e6`/`1e8` baselines used here are convenient (already fit, no new compute) but neither is parameter-matched to `confocal_random_zernike` as closely as `data/R80_N200_A1000000_..._f0.3000` would be (exact n_{atoms}/cloud-prior/shot-count match, already on disk, just never run through `generate_kinematic_estimates.py`) – worth doing if a tight, controlled aberration-only comparison is needed later.
- **Connect to the β (signal) CRB.** Propagate whatever aberrated-vs-Gaussian theta-CRB difference is established through `crb_signal.py`'s existing $j_{\text{eff}} \to$ beta-CRB machinery to get an actual quantitative bound on how much wavefront aberration/curvature costs in signal-extraction precision – the original motivating question for this whole line of work.
- The current check used only `run_000`'s first 10 shots for the CRB side (cheap: ~ 1 minute for all of §3–4 combined). Widening this to more shots/runs would tighten the CRB-side estimate too (currently it isn't the limiting source of noise in the comparison, but worth checking as the empirical run finishes all 20 runs).